

Benchmarking 62 Text Summarization Methods for Biomedical Literature

Benchmark Corpus



1,000 biomedical abstracts

20 journals · ScienceDirect and Cell Press

Author-written highlights
as reference summaries

Summarization Methods (n = 62)

62,000 generated summaries



Traditional
n = 2



Encoder-decoder
n = 11



Small LM
n = 28



Large LM
n = 21

general-purpose

domain-specific

reasoning-oriented

Multi-Metric Evaluation (n = 12)

LEXICAL (n = 5)

ROUGE-1 / -2 / -L
BLEU · METEOR

SEMANTIC (n = 3)

BERTScore
embedding similarity

FACTUAL (n = 4)

AlignScore · SummaC
MiniCheck



z-score normalization



Overall Performance Rank



Validated by blinded experts



Validated by an LLM-judge panel

Key Findings

- ✓ General-purpose models outperform domain-specific and reasoning-oriented models
- ✓ Medium-sized models outperform large-scale models
- ✓ Classical extractive methods lag behind neural approaches

✓ Mistral top-ranked family

✓ Judges match experts

✓ No data-leakage evidence